1. **Review of vectors :** Following vectors are given in the problem :

$$\mathbf{A} = \hat{x} + \hat{y} - \hat{z},$$
$$\mathbf{B} = (1, -2, -3),$$
$$\mathbf{C} = 2\hat{x} - \hat{y} + 3\hat{z}.$$

   a) (1 pt)$\mathbf{D} = \mathbf{A} + \mathbf{B} = 2\hat{x} - \hat{y} - 4\hat{z}$.

   b) (1 pt)$\mathbf{A} + \mathbf{B} - 4\mathbf{C} = 2\hat{x} - \hat{y} - 4\hat{z} - 4(2\hat{x} - \hat{y} + 3\hat{z}) = -6\hat{x} + 3\hat{y} - 16\hat{z}$.

   c) (2 pts)$|\mathbf{A} + \mathbf{B} - 4\mathbf{C}| = \sqrt{(-6)^2 + (3)^2 + (-16)^2} = \sqrt{301} \approx 17.35$.

   d) (2 pts)$\mathbf{A} + 2\mathbf{B} - \mathbf{C} = \hat{x} - 2y - 10\hat{z}$.
   The magnitude of the vector is $|\mathbf{A} + 2\mathbf{B} - \mathbf{C}| = \sqrt{1^2 + (-2)^2 + (-10)^2} = \sqrt{105}$. Then, the unit vector is written as

$$\hat{u} = \frac{\mathbf{A} + 2\mathbf{B} - \mathbf{C}}{|\mathbf{A} + 2\mathbf{B} - \mathbf{C}|} = \frac{\hat{x} - 2\hat{y} - 10\hat{z}}{\sqrt{105}} \approx 0.098\hat{x} - 0.195\hat{y} - 0.976\hat{z}.$$

   e) (2 pts)$\mathbf{A} \cdot \mathbf{B} = (\hat{x} + \hat{y} - \hat{z}) \cdot (\hat{x} - 2\hat{y} - 3\hat{z}) = (1 \times 1) + (1 \times (-2)) + ((-1) \times (-3)) = 2$.

   f) (2 pts)$\mathbf{B} \times \mathbf{C} = (\hat{x} - 2\hat{y} - 3\hat{z}) \times (2\hat{x} - \hat{y} + 3\hat{z}) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & -2 & -3 \\ 2 & -1 & 3 \end{vmatrix} = \hat{x}(-6 - 3) + \hat{y}(-6 - 3) + \hat{z}(-1 - (-4)) = -9\hat{x} - 9\hat{y} + 3\hat{z}$.

2. **Electrostatic charges and fields :**

   a) The electric fields at point P_3 and P_4 are the superposition of those induced by the charges Q_1 and Q_2. Referring to page 2 of Lecture 2, the field at point P_3 is given by(5 pts)

$$\mathbf{E}_3 = \sum_{n=1}^{2} \frac{Q_n}{4\pi\epsilon_o |\mathbf{r}_3 - \mathbf{r}_n|^2} \cdot \frac{\mathbf{r}_3 - \mathbf{r}_n}{|\mathbf{r}_3 - \mathbf{r}_n|} = 9\frac{\hat{z}}{|\hat{z}|^3} - 9\ \frac{(-\hat{z})}{|\hat{z}|^3} = 18\hat{z}\ \frac{\text{V}}{\text{m}},$$

   and the field at point P_4 is calculated as(5 pts)

$$\mathbf{E}_4 = \sum_{n=1}^{2} \frac{Q_n}{4\pi\epsilon_o |\mathbf{r}_4 - \mathbf{r}_n|^2} \cdot \frac{\mathbf{r}_4 - \mathbf{r}_n}{|\mathbf{r}_4 - \mathbf{r}_n|} = 9\frac{\hat{x} + \hat{z}}{|\hat{x} + \hat{z}|^3} - 9\ \frac{(\hat{x} - \hat{z})}{|\hat{x} - \hat{z}|^3} = \frac{9}{\sqrt{2}}\hat{z}\ \frac{\text{V}}{\text{m}}.$$

   b) (5 pts)**Bonus problem :** Plot of these fields in part (a) we obtained using Matlab is shown in the following figure.

```
eps0 = 8.85*10e-12; q = 10e-9;
r1 = [0, 0 , -1]; q1 = 1*q;
r2 = [0 , 0 , 1]; q2 =-1*q;

r3 = [0 , 0 , 0]; r4 = [1 , 0 , 0];

e31 = q1/(4*pi*eps0*mag(r3-r1).^3)*(r3-r1);

e32 = q2/(4*pi*eps0*mag(r3-r2).^3)*(r3-r2);

e3 = e31+e32;

e41 = q1/(4*pi*eps0*mag(r4-r1).^3)*(r4-r1);

e42 = q2/(4*pi*eps0*mag(r4-r2).^3)*(r4-r2);

e4 = e41+e42;
```

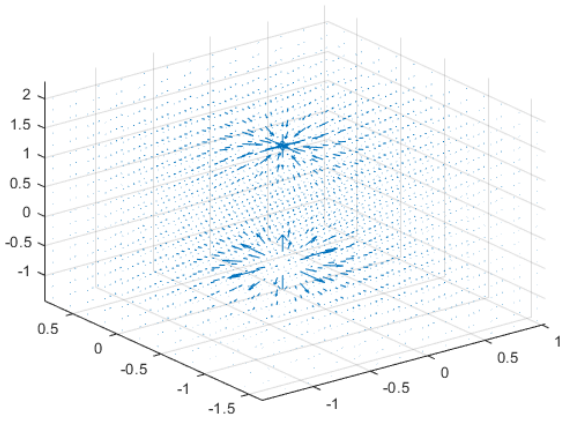
```
x = -2:0.2:2;
y = -2:0.2:2;
z = -2:0.2:2;
[X,Y,Z] = meshgrid(x,y,z);
e = zeros(21,21,21);

for ix= 1:21
  for iy = 1:21
    for iz = 1:21
      r = [X(ix,iy,iz) Y(ix,iy,iz) Z(ix,iy,iz)];
     e1 = q1/(4*pi*eps0*mag(r-r1).^3)*(r-r1);
     e2 = q2/(4*pi*eps0*mag(r-r2).^3)*(r-r2);
     e = e1+e2;
      eX(ix,iy,iz) = e(1);
      eY(ix,iy,iz) = e(2);
      eZ(ix,iy,iz) = e(3);

    end
   end
end

figure
quiver3(X,Y,Z,eX,eY,eZ,1.5);

function magnitude = mag(a)
magnitude = sqrt(sum(a.*a));
end
```



3. (10 pts)**Fields from forces - Lorentz force :** The fields $\mathbf{E}$ and $\mathbf{B}$ are the same for all three cases. However, the velocities of the particle for these cases are dfferent. Using the Lorentz force equation $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$, we write

$$\begin{aligned} 2\hat{z} &= q\mathbf{E}, \\ 0 &= q(\mathbf{E} + 2\hat{y} \times \mathbf{B}), \\ \hat{z} &= q(\mathbf{E} + -\hat{x} \times \mathbf{B}) \end{aligned}$$

for three cases. From these three equations, we obtain

$$\begin{aligned} \mathbf{E} &= \frac{1}{2}\hat{z}, \\ 2\hat{y} \times \mathbf{B} &= -\frac{1}{2}\hat{z} \\ -\hat{x} \times \mathbf{B} &= -\frac{1}{4}\hat{z}. \end{aligned}$$

Substituting $\mathbf{B}$ by $\mathbf{B} = \hat{x}B_x + \hat{y}B_y + \hat{z}B_z$, we write

$$\begin{aligned} 2\hat{y} \times (\hat{x}B_x + \hat{y}B_y + \hat{z}B_z) &= -\frac{1}{2}\hat{z} \implies -2B_x\hat{z} + 2B_z\hat{x} = -2\hat{z}, \\ -\hat{x} \times (\hat{x}B_x + \hat{y}B_y + \hat{z}B_z) &= -\frac{1}{4}\hat{z} \implies B_z\hat{y} - B_y\hat{z} = -\frac{1}{4}\hat{z}, \end{aligned}$$

from which we obtain $B_x = B_y = \frac{1}{4}$ and $B_z = 0$. Consequently, $\mathbf{E} = \frac{1}{2}\hat{z}\,\frac{\mathrm{V}}{\mathrm{m}}$, and$\mathbf{B} = \frac{1}{4}(\hat{x} + \hat{y})\,\frac{\mathrm{Wb}}{\mathrm{m}^2}$.

4. **Field transformations :**

   a) (5 pts)Since the Lorentz force equation has the same form in all inertial reference frames, $\mathbf{E}' + \mathbf{v}' \times \mathbf{B}' = \mathbf{E} + \mathbf{v} \times \mathbf{B}$. Because, the particle q is static in the reference frame, $\mathbf{v}' = 0$ in that reference frame. Then, the electric field measured in the frame of reference of q can be written as

$$\mathbf{E}' = 4\hat{z} + (2\hat{x} \times -2\hat{y}) = \mathbf{0}\,\frac{\mathrm{V}}{\mathrm{m}}.$$

   b) (5 pts)As $\mathbf{E}' = 0$, $\mathbf{F} = m\mathbf{a} = 0$. Therefore, the force experienced by the particle will be zero. In this case, the particle will not be accelareting.

5. *(10pts)* **Current density and surface integrals**

a) Let us use S_1, S_2, S_3, S_4, S_5 and S_6 to denote the surfaces $x=0$, $x=1$, $y=0$, $y=1$, $z=0$ and $z=1$, respectively. We consider S_1 and S_2 first. The unit vector on S_1 pointing away from the volume is along $-\hat{x}$ direction, so we have

$$\int_{S_1} \mathbf{J}\cdot d\mathbf{S} = \int_0^1\int_0^1 [-z^2(\hat{x}+\hat{y}+\hat{z})]\,|_{x=0}\cdot(-\hat{x})dydz = \frac{1}{3}\,(A).$$

For S_2, the unit vector pointing away from the volume is along $+\hat{x}$ direction, therefore

$$\int_{S_2} \mathbf{J}\cdot d\mathbf{S} = \int_0^1\int_0^1 [-z^2(\hat{x}+\hat{y}+\hat{z})]\,|_{x=1}\cdot(+\hat{x})dydz = -\frac{1}{3}\,(A).$$

Similarly, for S_3, S_4, S_5 and S_6, we can obtain

$$\int_{S_3} \mathbf{J}\cdot d\mathbf{S} = \int_0^1\int_0^1 [-z^2(\hat{x}+\hat{y}+\hat{z})]\,|_{y=0}\cdot(-\hat{y})dxdz = \frac{1}{3}\,(A),$$

$$\int_{S_4} \mathbf{J}\cdot d\mathbf{S} = \int_0^1\int_0^1 [-z^2(\hat{x}+\hat{y}+\hat{z})]\,|_{y=1}\cdot(+\hat{y})dxdz = -\frac{1}{3}\,(A),$$

$$\int_{S_5} \mathbf{J}\cdot d\mathbf{S} = \int_0^1\int_0^1 [-z^2(\hat{x}+\hat{y}+\hat{z})]\,|_{z=0}\cdot(-\hat{z})dxdy = 0\,(A),$$

$$\int_{S_6} \mathbf{J}\cdot d\mathbf{S} = \int_0^1\int_0^1 [-z^2(\hat{x}+\hat{y}+\hat{z})]\,|_{z=1}\cdot(+\hat{z})dxdy = -1\,(A).$$

Therefore,

$$\oint_S \mathbf{J}\cdot d\mathbf{S} = \sum_{i=1}^{6}\int_{s_i} \mathbf{J}\cdot d\mathbf{S} = -1\,(A).$$

b) Divergence theorem followed by Continuity equation:

$$\oint_S \mathbf{J}\cdot d\mathbf{S} = \int_V \nabla\cdot\mathbf{J}dV = -\frac{\partial}{\partial t}\int_V \rho dV = -\frac{\partial Q_{enclosed}}{\partial t}$$

In words, since the current density flux is negative, there is current going into the charged region, thus due to charge conservation, we should expect an increase (gain) of charges within it.

c) By Gauss law: *Electric flux* $\sim$ *Charges enclosed*
From (b), an increase of charge would lead to an increase of eletric flux through the enclosed surface.